

S1 Table

Highlights from this week's report:

- Influenza severity for Massachusetts is low this week.
- The percent of influenza-like illness (ILI) visits in Massachusetts is 0.61%, which is lower than the regional baseline of 2.0%.
- The percent of hospitalizations associated with influenza is 0.08%, which is lower than two of the previous seasons, but comparable to last season in the same week.
- Overall ILI activity is minimal. All regions are reporting minimal ILI activity.
- More influenza A than influenza B positive specimens have been reported by hospitals and outpatient facilities in Massachusetts. For influenza A, the predominant strain is currently H3N2.
- Nationally, influenza activity is increasing in most of the country.
- Additional statewide and national data including geographic spread, ILI activity, and pneumonia and influenza mortality are available at CDC's FluView Weekly Report at www.cdc.gov/flu/weekly and FluView Interactive <https://www.cdc.gov/flu/weekly/fluviewinteractive.htm>.
- Statewide and national COVID data are available at <https://www.mass.gov/info-details/covid-19-response-reporting> and <https://www.cdc.gov/coronavirus/2019-ncov/covid-data/covidview/index.html>

It's not too late to get vaccinated. Flu vaccination is always the best way to prevent flu and its potentially serious complications.

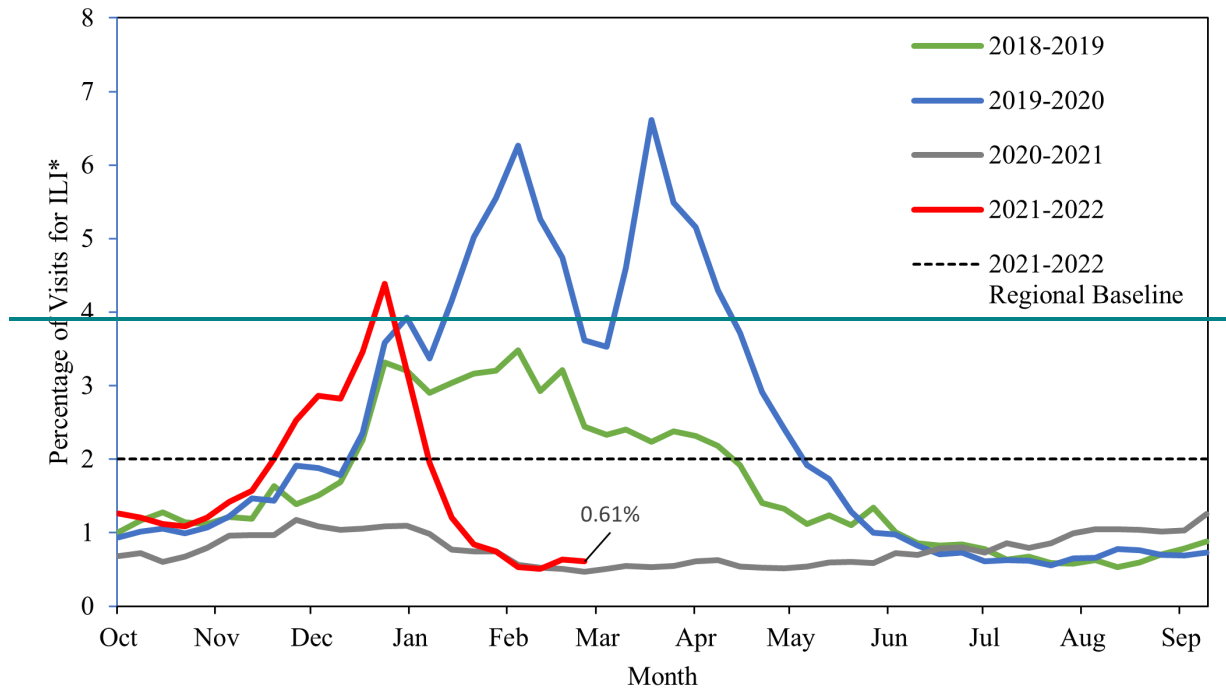
Influenza-like illness activity

Influenza-like illness (ILI) is defined as fever (temperature of 100 deg F or greater) in addition to cough and/or sore throat. Many more people are infected with influenza than are tested for influenza. ILI is used throughout the regular influenza season to help track influenza activity in individuals who are not tested, as trends in ILI have been shown to mirror influenza trends. Ninety three healthcare facilities called 'sentinel sites' report the number of patients they see with ILI each week during regular flu season to the Massachusetts Department of Public Health (MDPH). Sentinel sites include provider offices, school health services, community health centers, urgent care centers, and emergency departments across Massachusetts. Data reported by emergency departments provide information about ED visits that include diagnostic codes (influenza diagnosis code) as well as terms indicative of ILI. The CDC uses trends from past years to determine a region-specific baseline rate of ILI visits, which for Massachusetts is 2.0%. A rate above this

regional baseline indicates higher than normal levels of ILI in the state. For more information on how regional baselines are calculated see CDC's influenza surveillance website at <https://www.cdc.gov/flu/weekly/overview.htm>.

Figure 1 shows that the percent of ILI visits in the current week is lower than the regional baseline.

**Figure 1. Percentage of Visits for Influenza-Like Illness (ILI) Reported by Sentinel Provider Sites in Massachusetts
October 3, 2021 - March 5, 2022**



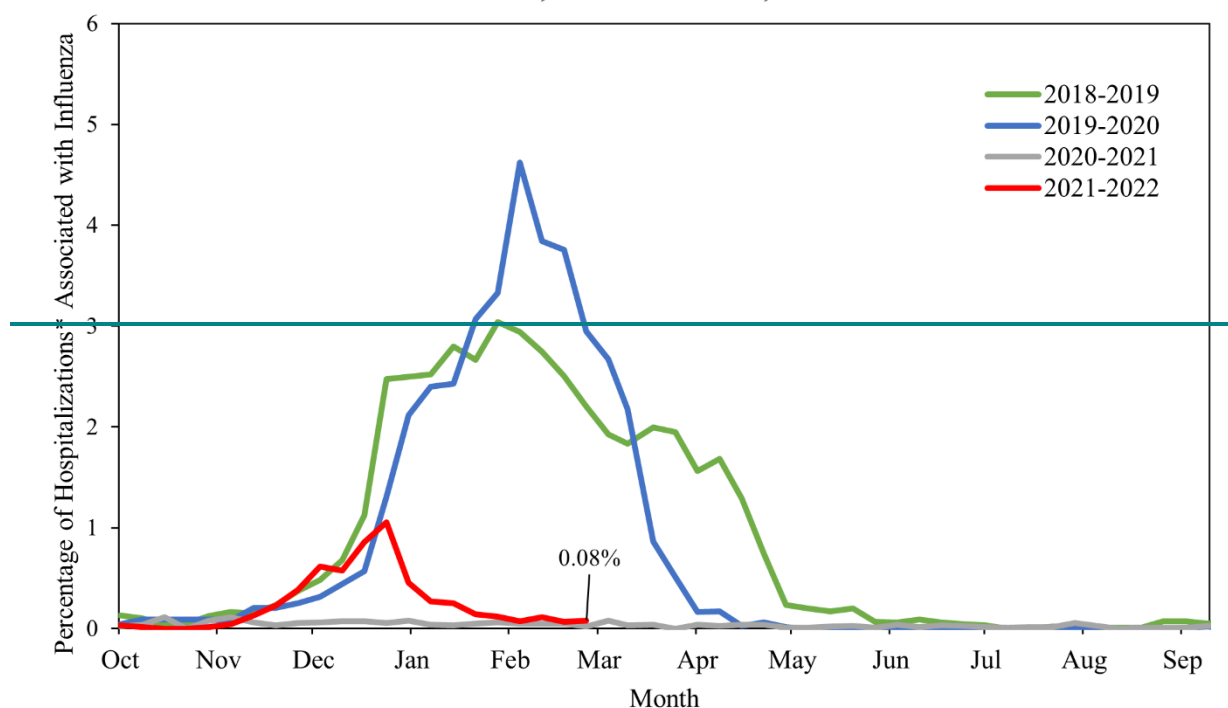
*Influenza-like illness (ILI, defined by fever $\geq 100^{\circ}\text{F}$ and cough and/or sore throat), as reported by Massachusetts sentinel surveillance sites. ILI reported by sentinel sites which report via ED syndromic surveillance include cases meeting the ILI definition and cases with a diagnosis indicating influenza infection. The 2021-2022 regional baseline is 2.0%, for more information on how this baseline is calculated please visit <https://www.cdc.gov/flu/weekly/overview.htm>.

Influenza-associated hospitalizations

As part of the National Syndromic Surveillance Program, MDPH receives data from Emergency Departments (EDs) covering 100% of ED visits statewide. These data are used to track patient visits related to influenza by monitoring the diagnoses the patients receive (ICD-10 code). These data are available to MDPH in near real time.

Figure 2 shows the percent of all ED visits which result in a patient hospitalized because of illness associated with influenza infection. The percentage of influenza-associated hospitalizations is lower than two of the previous seasons, but comparable to last season in the same week.

Figure 2. Percentage of Hospitalizations Associated with Influenza in Massachusetts
October 3, 2021 - March 5, 2022

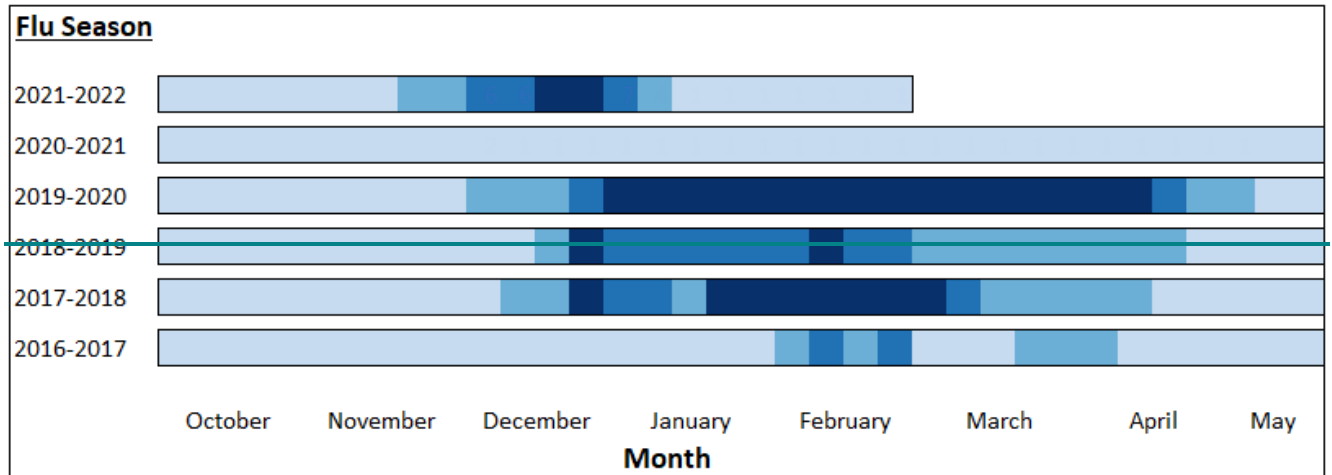


*All patients admitted through hospital emergency departments as captured by syndromic surveillance

ILI Activity in Massachusetts

Figure 3 shows the current season's weekly ILI activity compared to the last five flu seasons in Massachusetts. ILI activity for each week is categorized as minimal, low, moderate, or high, with a shade of blue corresponding to the category of ILI activity for that week. Darker shades of blue indicate more intense ILI activity. Figure 3 shows that ILI activity in Massachusetts is minimal this week.

Figure 3. ILI Activity in Massachusetts Reported Weekly by Sentinel Sites



Variable	Tumor sample	Plasma/ serum	RNA
<u>Ang-2</u>	<u>Zhang L., 2003 [1]:</u> 52 specimen Result: Expression in endothelium and 12% of tumor cells	<u>Sallinen H., 2010 [5]:</u> 150 patients Result: Higher pre-operative Ang-2 levels in cancer. Poor PFS and OS.	<u>Hata K., 2004 [9]:</u> 85 specimens Result: Low Ang-1/ Ang-2 predicted poor OS (univariate analysis)
	<u>Lin Z., 2011 [2]:</u> 21 patients Result: No correlation with clinicopathological parameters, no difference between cancer vs benign	<u>Sallinen H., 2014 [6]:</u> 75 patients Result: Higher Ang-2 and Ang-2/ VEGFR2 in cancer. Poor PFS and OS.	

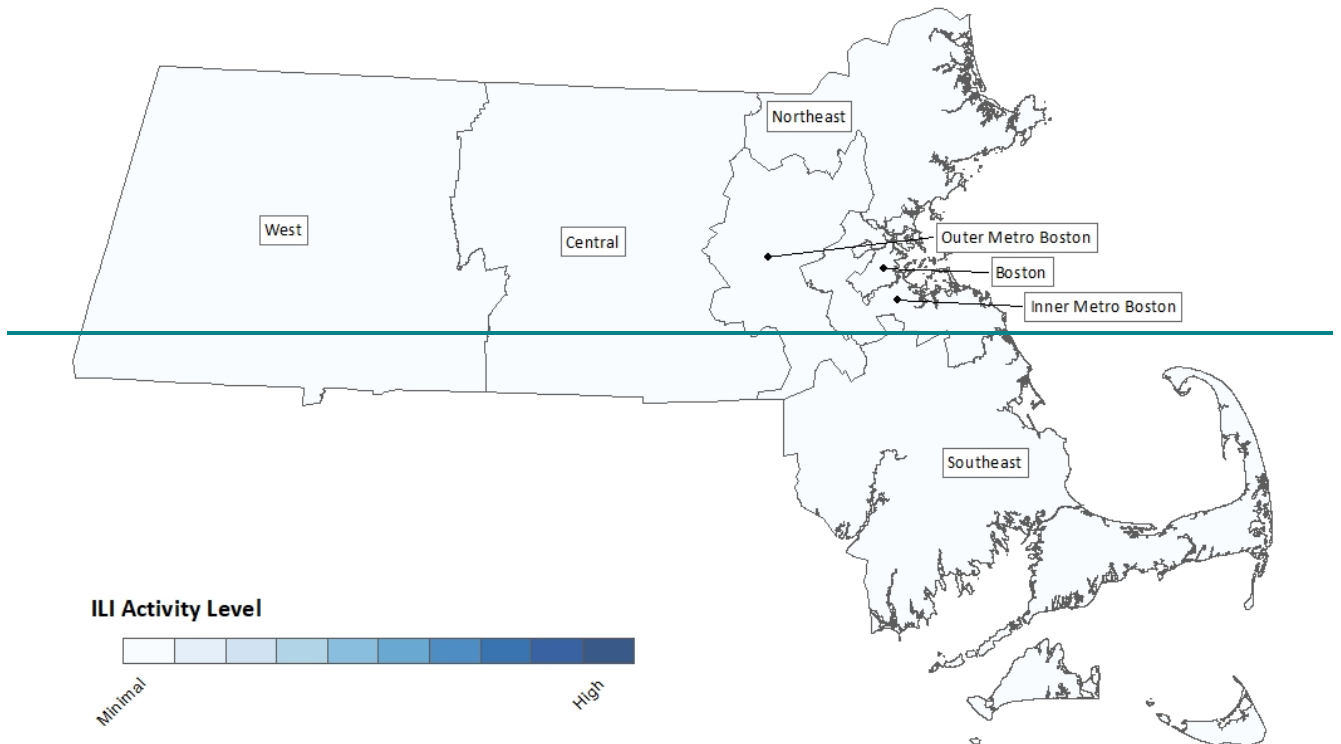
<u>Tie-2/ Ang-2</u> Minimal	<u>Li S., 2010 [3]:</u> 12 specimens Result: Cytoplasmic expression, higher expression in cancer, no correlation with clinical parameters Low	<u>Backen A., Clin Cancer Res 2014 [7]:</u> 91 patients Result: Moderate High Ang-1/Tie-2 improved PFS in bevacizumab treated patients	<u>HighHata K., Oncology 2002 [10]:</u> 19 patients Result: No difference with Ang-2 and Tie-2 expression in cancer vs benign tumors
	<u>Brunckhorst M.K., 2014 [4]:</u> 12 cancer and 12 omental metastatic samples Result: Ang-2 expression in tumor cells, Tie-2 in endothelium and fibroblasts. Higher Ang-2 expression in cancer.	<u>Zhou C., 2016 [8]:</u> 92 patients Result: 50% increase of Tie-2 from the nadir predicted tumor progression after bevacizumab	

S2 Table

ILI Activity in Massachusetts by Region

Figure 4 shows the relative intensity of reported ILI activity in Massachusetts by region. Although regions may not all experience the same intensity of ILI at similar times, infections due to influenza can be found throughout Massachusetts during flu season. Figure 4 shows that all regions are reporting minimal ILI activity.

Figure 4: ILI Activity Reported Weekly by Massachusetts Sentinel Sites

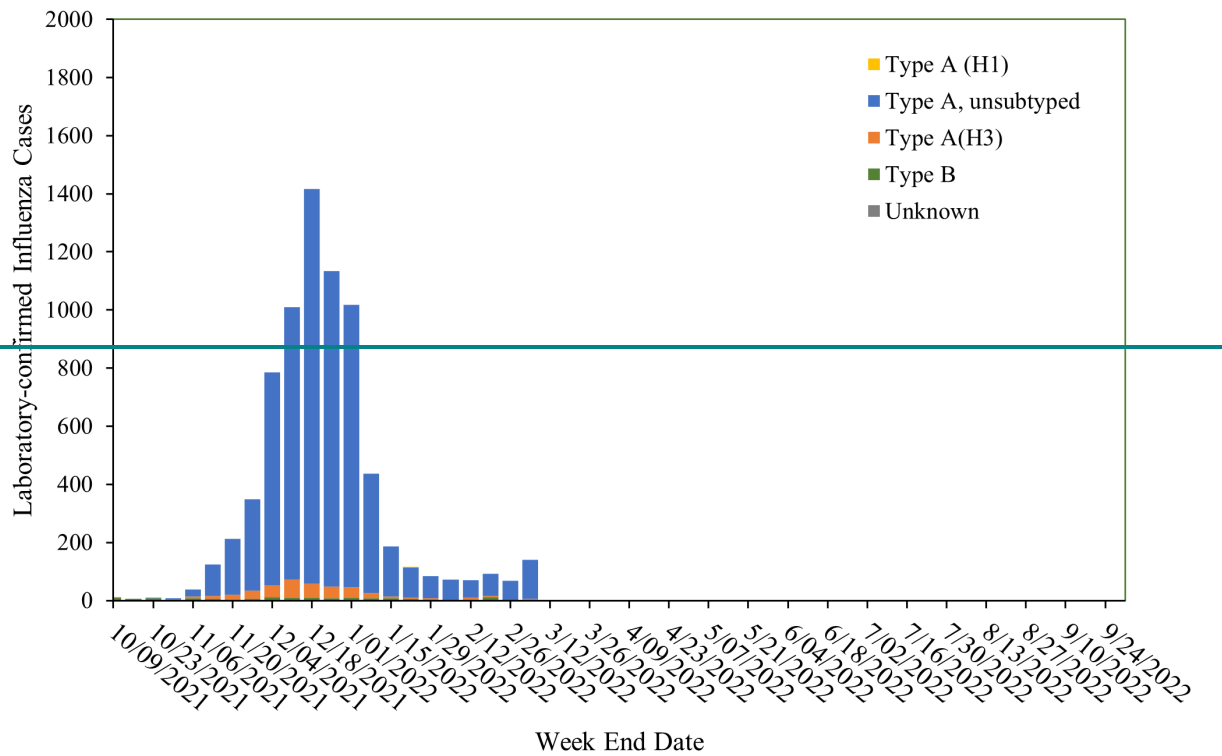


Laboratory testing for influenza

Laboratories in Massachusetts report all positive influenza test results to MDPH. The majority of individuals with influenza-like illness are not tested; therefore, the number of positive test results does **not** reflect the total number of influenza cases in Massachusetts. However, laboratory data do provide information about the types of influenza virus circulating in Massachusetts and help indicate the presence and define the distribution of influenza in the state.

Figure 5 illustrates the number of laboratory-confirmed influenza cases in Massachusetts by week. More influenza-A than influenza-B positive specimens have been reported by hospitals and outpatient facilities in Massachusetts.

**Figure 5. Laboratory-confirmed Influenza Cases in Massachusetts
October 3, 2021 - March 5, 2022**



*Influenza cases confirmed via PCR or viral culture test by specimen collection date.

Testing at the State Public Health Laboratory

The Massachusetts State Public Health Laboratory (MA SPHL) performs influenza surveillance testing year-round to confirm circulating influenza virus types. Samples are submitted by outpatient healthcare providers (ILINet) and hospital diagnostic laboratories in Massachusetts. For the 2021-2022 season, Table 1 summarizes the influenza surveillance testing conducted by MA SPHL beginning October 3, 2021. In the 2021-2022 flu season, 116 cases of seasonal A/H3N2 influenza and 1 case of B Victoria has been confirmed among 146 samples tested.

Table 1. Weekly Summary of Massachusetts State Public Health Laboratory Influenza Surveillance Test Results

Variable				Correlation coefficient (r)	p				
2021-2022 Season: Influenza Surveillance									
MMWR Week: (Specimen Collected)	2009 H1N1	seasonal A/H3N2	H3N2	B-Yam	B-Vic	N	U	T	T
						e	n	e	e
						-	n	e	e
						F	s	t	a
						t	a	t	R
						u	t	t	e
						p		T	e
						e		e	e
						s		s	'
						t		t	d
						%		e	
						}		d	
06 (02/06 - 02/12/22)	0	Ang-2 IRS	-0.64	-0.002	0	2	0	2	2
						{			
						±			
						0			
						0			
						%			
						}			
Ang-2 IRS* (07 (02/13 - 02/19/22)	-0.79	0	0	0	0	0	0	0	0
						±			
						0			
						0			
						4			
						{			
						0			
						%			
						}			
Ang-2 IRS (HGSC) (08 (02/20 - 02/26/22)	-0.87	≤0.001	0	0	0	0	0	0	0
						{			
						0			
						%			
						}			
Ang-2 met PP				0.75		0.005			
09 (02/27 - 03/05/22)	0	Tie-1 IRS	-0.008	-0.972	0	±	0	±	±
						{			
						±			
						0			
						0			
						%			
						}			

<u>Tie-1</u>	=	0	0	0	3	0	3	3
<u>IRS</u>	<u>0.0</u>	<u>0.7</u>			{			
<u>(* / HGS</u>	<u>87</u>	<u>99</u>			1			
<u>C) Prior 4</u>		3			0			
<u>wk Total</u>					0			
					%			
					}			
<u>Cumulative</u>	0	116	0	0	<u>Ti</u>		1	1
<u>e-Season</u>					<u>e-</u>	0	0	4
<u>total</u>					<u>1</u>		6	5
					<u>m</u>	2	5	
					<u>et</u>	1	0	
					<u>I</u>	5	2	
					<u>R</u>	1	7	
					<u>S</u>	1		
					7			
					{			
					8			
					0			
					%			
					}			
<u>Tie-2 IRS</u>			<u>0.163</u>		<u>0.479</u>			
<u>Tie-2 IRS</u>			<u>-0.319</u>		<u>0.339</u>			
<u>(* / HGSC)</u>								
<u>Tie-2 met IRS</u>			<u>0.211</u>		<u>0.510</u>			

p-values determined by Spearman's test, IRS immunoreactive score, HGSC high grade serous carcinoma, Ang-2 met angiopoietin-2 metastasis, PP percentage of positively stained epithelial cells, * serous histologic type

S3 Table

All data are subject to change as test results become finalized. The 2021-2022 influenza season began the week of 10/03-10/09/2021.

All specimens which test negative for influenza at MA SPHL are also tested for non-influenza respiratory diseases including COVID-19 (SARS-CoV-2), respiratory syncytial virus (RSV), rhinovirus (RV)/enterovirus (EV), parainfluenza virus (PIV), human metapneumovirus (HMPV), seasonal human coronavirus (HCV) and adenovirus (ADENO). HCV does not include COVID-19. In the 2021-2022 flu season, 1 case of RV/EV, 1 case of PIV, 1 case of HMPV, 1 case of ADENO and 1 co-infection has been confirmed among 102 samples tested.

Table 2. Weekly Summary of Massachusetts State Public Health Laboratory non-Influenza Respiratory Surveillance Test Results

2021-2022 Season: Influenza-like Illness Surveillance	Ang2					p	Tie1					p	Tie2					p																	
Grade																																			
Low																																			
IRS	8										3					2																			
PP	80										65					40																			
High																																			
IRSMM WR Week: (Specimen Collected)	6 SARS- CoV- 2	0.075			2 RSV	0.018			2 PIV	2 HMPV	0.108			HCV	A D E N O	Co- infe- ctio- n*	No. Pos (%)	Unsat	Tot al Test ed	T o t a l R e- e- d															
PP	65										0.008					25					0.007					30					0.016				
Stage																																			
I-II																																			
IRS	9										2					2					2														
PP	75										20					40																			
III-IV																																			
IRS06 (02/06 - 02/12/2 2)	60	0	0	0	0	0	0	0	0	0	0	0	0	0.185 (0.047 *)%	2	0.781	2	0	1 4 1 2 0																
PP07 (02/13 - 02/19/2 2)	65 0	0.370			3 50	0.657			35 0	0	0	0	0.375(0%)	0	0	0	0	0																	
Histo logy	0	0	0	0	0	0	0	0	0	0	0	0	0(0%)	0	0	0	0	0																	
08 (02/20 - 02/26/2 2)																																			

	0	0	0	0	0	0	0	0(0%)	0	1	1
<i>Serous</i>											
IRS	6	0.436	2	0.002	2	0.002					
PP	65	0.110	40	0.002	30	<0.001					
<i>Mucinous</i>	60	0.867	30	0.029	60	0.001	0	0	0(0%)	0	3
IRS ^{Prior}											
4-wk Total											
PP	75	0.255	60	0.024	75	<0.001					
<i>Endometrioid</i>											
IRS	6	0.175	2	0.411	2	0.672					
PP	65	0.878	20	0.136	33	0.890					
<i>Clear cell</i>	40	0.477	1	0.001	1	0.095	1	1	3(3%)	0	102
IRS ^{Cum}											1
ulative											0
Season											2
total											
PP	30	0.020	5	0.003	10	0.024					
Residual tumor											
<i>None</i>											
IRS	6		2.5		3						
PP	70		45		40						
<i>≤1cm</i>											
IRS	6	0.109	2.5	0.583	2	0.923					
PP	65	0.746	45	0.496	30	0.854					
<i>>1cm</i>											
IRS	6	0.012	2	0.465 (0.008*)	2	0.688					
PP	60	0.092	25	0.284 (0.020*)	30	0.526					
Recurrence											
<i>No</i>											
IRS	6		3		3						
PP	73		45		42.5						
<i>Yes</i>											
IRS	6	0.018	2	0.157 (0.025*)	2	0.202					
PP	63	0.006	25	0.091 (0.009*)	30	0.137					

p values determined by Kruskal-Wallis test followed by Mann-Witney tests when appropriate; *p value reported for serous tumors in brackets when only comparison for this subgroup was statistically significant; IRS, PP are median values; HGSC = high grade serous cancer, IRS immunoreactive score, PP percentage of positively stained cells, Recurrence = recurrence of cancer

References

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~~All data are subject to change as test results become finalized. The 2021–2022 influenza season began the week of 10/03–10/09/2021.~~

~~*Coinfection is the simultaneous detection of two or more of the non-influenza respiratory diseases included in this table.~~

~~MA SPHL submits a subset of influenza samples to CDC for further genetic analysis (antigenic characterization).~~

~~Every two weeks MA SPHL screens influenza specimens to detect mutations within influenza A/H3N2 and A/2009 H1N1 viruses to look for antiviral resistance once positive specimens have been identified.~~

~~Additional information on national antiviral resistance testing including recommendations for antiviral treatment and chemoprophylaxis of influenza virus infection can be found at <http://www.cdc.gov/flu/weekly/>.~~